Computer Science & IT



Fundamental of C Language

Data Types & Operator

Lecture No. 02



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Recap of Previous Lecture



Topic — Numbers System

Topic — Singed & unsigned

Topic — Binay, octal, Hexadecimal

Topic

Topic

Slide

Topics to be Covered



Topic	Structure of C program
Topic	Header file, main function
Topic	block, Declaration, Assignment
Topic	Input output statement
Topic	

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Structure of C program

```
#include<stdio.h>
int main() {
    int a=10, b=20;
    int c;
    c= a+b;
    printf("%d", c);
    return 0;
}
```

#include : preprocessor directive

Include a file in the program

angled bracket

stdio.h - standard input output file

→ header file



Structure of C program

```
#include<stdio.h>
int main() {
    int a=10, b=20;
    int c;
    c= a+b;
    printf("%d", c);
    return 0;
}
```

← # standard Header file included in program to perform input output operation.

Who will do ∴ preprocessor

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Structure of C program

```
#include<stdio.h>
int main() {
    int a=10, b=20;
    int c;
    c= a+b;
    printf("%d", c);
    return 0;
}
```

$f(x) = x^2$ $f(1) = 1$

int main ( ) ← main function

When program will execute then main function is called

int main() ← Mathematical function define

int main() → Name of function

outcome of function execution

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Structure of C program

```
#include<stdio.h>
int main() {
    int a=10, b=20;
    int c;
    c= a+b;
    printf("%d", c);
    return 0;
}
```

{ } > grouping of statement for a function

{ } > It creates a block.

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Structure of C program

```
#include<stdio.h>
int main() {
    int a=10, b=20;
    int c;
    c= a+b;
    printf("%d", c);
    return 0;
}
```

{ } → Scope & visibility of a variable

Whenever block is created Scope & visibility of variable is defined

a, & b can be accessed & modified within the block.

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Structure of C program

```
#include<stdio.h>
int main(){
    int a=10, b=20;
    int c;
    c= a+b;
    printf("%d", c);
    return 0;
}
```

Statement : In C Language after every statement ";" semicolon is present.

int - Data type

Name of variable / Identifier

= is an operator

10, 20 are Numeric constant

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Structure of C program

```
#include<stdio.h>
int main() {
    int a=10, b=20;
    int c;
    c = a+b;
    printf("%d", c);
    return 0;
}
```

Declarative statement

A statement which defines data type & Name of variable is declarative statement.

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Structure of C program

```
#include<stdio.h>
int main() {
    int a=10, b=20;
    int c;
    c= a+b;
    printf("%d", c);
    return 0;
}
```

Declarative & Initialization

If in a statement data type, Name of variable and initial value is given then we call that statement as Declarative & initialization.

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Structure of C program

```
#include<stdio.h>
int main() {
    int a=10, b=20;
    int c;
    c= a+b;
    printf("%d", c);
    return 0;
}
```

Arithmetic statement

A statement arithmetic operation is performed

c, a, b are variable

=, + are Arithmetic operator

C:20 ← C value is computed

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Structure of C program

```
#include<stdio.h>
int main() {
    int a=10, b=20; // declarative & Initialization
    int c;  //  Declarative statement
    c = a+b;  // Arithmetic statement
    printf("%d", c);
    return 0;
}
```

Comments

Comment are removed. it is used to make logic more readble.

```
// Single Line comment

/* stmt1.
stmt2;
stmt3; */
```

↑ multiline Comment

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Structure of C program

```
#include<stdio.h>
int main() {
    int a=10, b=20;
    int c;
    c = a+b;
    printf("%d", c);
    return 0;
}
```

Input/output Statement

Giving input to computer and taking output.

printf

↑ print function ← Declare in Header file

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Structure of C program

```
#include<stdio.h>
int main() {
    int a=10, b=20;
    int c;
    c= a+b;
    printf("%d", c);
    return 0;
}
```

f(x) , f(x,y)

printf("%d", c);

"%d" } argument of function
c }

%d → format specifier for integer

c is Name of variable



Structure of C program

```
#include<stdio.h>
int main() {
    int a=10, b=20;
    int c;
    c= a+b;
    printf("%d", c);
    return 0;
}
```

program execute Successfully

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Structure of C program

Types of statement

1. Declarative (Inihlization)
2. Anthmetic & logical (Relahim, bit wise)
3. Input out put statement
4. Control statement

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Structure of C program

```
#include<stdio.h>
int main() {
    int a=10, b=20;
    int c;
    c= a+b;
    printf("%d", c);
    return 0;
}
```

- Flow of execution
  - Sequential manner

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2 mins Summary

Topic

Header file, main funcha

Topic

block, Declarative statement

Topic

Arithmetic. Input output statement;

Topic

Topic

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THANK - YOU

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